Can the business survive and repay its bonds?

Creditworthiness, AI borrowing and the evidence behind the numbers

Corrected research & teaching edition · 6 November 2026 · Singapore time
Historical annual observations are labelled individually. No live market-price snapshot.

US corporate credit | AI infrastructure | Apple / SEC 10-K | Evidence coverage: partial

01 · Cash first

Repayment depends on cash and timing, not profit headlines. Apple's FY2026 operating cash flow less cash PP&E was US$136.056bn; it is not cash available after every obligation.

02 · Test the funding gap

AI investment can outrun internally generated cash. Oracle's FY2026 cash PP&E exceeded operating cash flow by US$23.686bn. Financing access and payment timing are therefore central.

03 · Read the claim

Creditors rank ahead of equity, but bondholders do not all rank equally. Collateral, issuer entity, guarantees and other priority claims change recovery.

04 · Price separately

A resilient issuer can still offer a poorly priced bond. Compare matched instruments and distinguish Treasury-rate, credit-spread, liquidity and FX risks.

Business — Where cash comes from → **Obligations** — What must be paid, when → **Claim** — Who gets paid first → **Price** — What risk you are paid to bear

Evidence boundary. Nine annual financial snapshots, audited teaching calculations and partial primary issuance coverage. No unsupported credit ratings, default forecasts, target prices or composite safety scores. The original broad peer universe and every new-issue field are not fully verified.

Sources and dates AAPL (2026-11-03) · ORCL (2026-06-22) · SEC_PRIORITY (Undated; checked 2026-11-06) · FINRA_SPREAD (Page checked 2026-11-06)

Your route through the evidence

Clickable contents · start with the question you need to answer

01 Can the business survive and repay its bonds?

03 Correct the framework before using the numbers

04 Default is only one way a bond can disappoint

05 Six questions turn accounts into a credit view

06 The legal claim can matter as much as the company

07 Use Morningstar to navigate, SEC filings to verify

08 Apple's FY2026 cash generation is large, but definitions still matter

09 Apple retains US$33.945bn after the selected cash uses

10 Apple can show net debt or net cash, depending on the definition

11 Eight reported annual snapshots cover cash investment; one does not

12 Cash investment absorbs very different shares of operating cash flow

13 A completed diagnostic scorecard is more useful than a fabricated rating

14 Accounting definitions can change the apparent winner

15 Trace who buys, who builds and who carries the obligations

16 AI credit risk follows contracts, not just company logos

17 Oracle's maturity profile is long, but its cash investment still needs funding

18 Original pricing must not masquerade as a current quote

19 A wider spread is meaningful only after the comparison is controlled

20 A solvent issuer can still produce a negative bond return

21 The corrected waterfall pays both claims in full at US$100

22 Interest coverage can remain positive while liquidity fails

23 Use three separate decisions before buying a bond

24 What was checked, what remains open

25 Sources and evidence trail 1

26 Sources and evidence trail 2

27 Sources and evidence trail 3

28 Sources and evidence trail 4

Read this as a sequence, not a league table

The first question is whether cash can meet obligations. Only then ask what your particular bond could recover and whether its market price compensates you. The Excel model keeps assumptions separate from reported inputs.

In everyday terms

A successful shop, a reliable borrower and a good investment are three different judgments. This report keeps them separate.

Correct the framework before using the numbers

This edition supersedes the unsupported statements in the earlier draft

Earlier problem	Correction used here
Debt described as below equity	Creditors are paid before common equity, subject to applicable legal priorities and claim structures.
$100 assets / $80 senior / $20 junior treated as a junior loss	$80 + $20 = $100. With no costs or other claims, both classes recover 100%.
EBIT described as EBITDA	Operating income is an EBIT proxy. An operating EBITDA proxy adds D&A and is not automatically covenant EBITDA.
Buybacks included in debt	Buybacks are cash distributions to shareholders, not borrowing balances.
Unsupported ratings and implied issuer rankings	No issuer rating is included without dated agency verification; no weighted safety score is invented.
Spread treated as default probability	Spreads include risk premia and instrument/market differences. Any PD illustration states its model assumptions.
Annual periods and cash-flow definitions mixed	Each fiscal year end is shown; gross cash investment, leases and company-defined FCF are separated.

Checks that can be reproduced

The workbook contains conservation and claim-cap tests. Each recovery scenario reconciles gross assets to costs, creditor payments and equity residual. Scenarios show operating cash flow, principal payments and minimum-cash needs separately.

Sources and dates SEC_PRIORITY (Undated; checked 2026-11-06) · NYFED (2026-02-24)

Default is only one way a bond can disappoint

Creditworthiness is the capacity and willingness to pay the promised cash flows

Risk	What can go wrong	What to inspect
Issuer default	Interest or principal is missed, or debt is restructured under distressed conditions.	Cash generation, liquidity, payment schedule, covenant terms.
Interest rates	The benchmark yield rises and the bond's price falls even if repayment remains credible.	Duration, maturity, coupon and holding horizon.
Credit spread	Required compensation above the benchmark increases.	Issuer news, peer spreads, supply, risk appetite, liquidity.
Trading liquidity	A sale requires a discount or a wide bid-offer spread.	Issue size, trade frequency, dealer quotes and settlement.
FX and optionality	Currency moves reduce home-currency returns; calls change cash-flow timing.	Currency exposure, hedges, call schedule, yield-to-worst.
Valuation	You pay too much for the risks and contractual cash flows.	Comparable instruments, market price, fees and alternatives.

Default risk versus loss

Default probability and loss severity are different. For a common horizon, expected credit loss = exposure at default × probability of default × loss given default. A legal recovery paid years later also has a time value.

Ratings are opinions, not insurance

A credit rating is not a price target or a guarantee. Agency methods, incentives, timing and issue-specific protections matter. Ratings do not remove market-price or liquidity risk. This report omits unverified issuer ratings.

In everyday terms

The borrower can keep paying while your bond's screen price falls. A promise being honoured and a price staying still are not the same thing.

Sources and dates SEC_BOND (Undated; checked 2026-11-06) · SEC_RATING (2017-10-12) · FINRA_YIELD (Page checked 2026-11-06) · NYFED (2026-02-24)

Six questions turn accounts into a credit view

An original teaching framework, not a Morningstar or rating-agency methodology

Earn Durable demand? → **Convert** Profit into cash? → **Fund** Cash after investment? → **Refinance** Maturities covered? → **Recover** Claim and collateral? → **Price** Compensation sufficient?

Question	Calculation / evidence	Interpretation and limit
Is the business durable?	Segments, customer concentration, competition, regulation, supplier dependence.	A moat can support future cash flow, but does not establish a credit rating.
Does profit become cash?	OCF / net income; working capital; taxes; stock compensation.	One year can be distorted by timing. OCF is not the same as recurring cash earnings.
Can investment be funded?	Cash surplus = OCF − cash capex.	Add finance-lease principal, other investment and distributions as relevant.
How much leverage?	Borrowings / EBITDA; net debt = specified debt − specified liquid assets.	Use consistent definitions. Do not treat all liabilities as borrowings or all investments as cash.
Can interest and maturities be paid?	EBIT / gross interest; sources / uses over the next 12–24 months.	Interest coverage is not principal coverage. Undrawn lines may be conditional.
What changes in stress?	Revenue × margin → cash → debt service → closing liquidity.	Test cash buffers and refinancing assumptions, not only accounting ratios.

No single ratio calculates “survivability”

A defensible probability of default requires a calibrated model. Balance-sheet ratios, market signals and qualitative evidence can inform a view; they do not create a precise probability without a defined horizon, historical calibration and validation. A mechanical score is a screening device, not a certified rating.

In everyday terms

Look at the shop’s earning power, then its bank account, then the bills due. A profitable year does not settle a bill that arrives before the cash.

Sources and dates AAPL (2026-11-03) · ORCL (2026-06-22) · ORCL_DEBT (2026-06-22) · SEC_BOND (Undated; checked 2026-11-06) · SEC_RATING (2017-10-12) · NYFED (2026-02-24)

The legal claim can matter as much as the company

General teaching framework; actual rights depend on the instrument and applicable law

Legal entity — Who owes the money? → **Collateral** — Which assets support it? → **Priority** — Which claims rank ahead? → **Covenants** — What can change or accelerate?

Structure	Plain-English meaning	What to check
Secured debt	Rights over specified collateral, rather than a general promise alone.	Collateral ownership, valuation, lien priority, other liens, enforcement costs and shortfall treatment.
Senior unsecured debt	A general claim without specified collateral; senior within the contractual ranking stated.	Other secured or priority claims, guarantees and the legal issuer. Senior does not mean no losses.
Subordinated debt	Contractually ranks behind defined senior claims.	Exactly which claims rank ahead, payment-blocking provisions and any deferral or conversion terms.
Parent versus subsidiary	Creditors of operating subsidiaries can be structurally ahead of a parent-only bond.	Which entity owns the assets and earns the cash; upstream dividend restrictions and guarantees.
Equity	Residual ownership after creditor claims in the simple waterfall.	Equity value is not the same as recoverable assets available to bondholders.

Covenants are contractual promises and restrictions

Read the definition, not just the headline ratio. Covenants can require reporting, restrict liens or asset transfers, and impose financial tests. Cure periods, exceptions, waivers and acceleration rights matter. A covenant breach is not automatically the same event as a missed cash payment. Oracle’s defined coverage test, for example, is not the simple EBIT/gross-interest ratio in this report.

A guarantee can change where a bondholder has a claim. A famous group name is not a substitute for checking the issuer, guarantor, ranking and enforceable documents. Do not apply a single simplified priority diagram to every jurisdiction or capital structure.

> **In everyday terms**
> Two people can lend to the same group and hold different promises. One may have security over an asset; another may rely on whatever reaches the parent after other creditors are paid.

Sources and dates SEC_BOND (Undated; checked 2026-11-06) · SEC_PRIORITY (Undated; checked 2026-11-06) · ORCL_DEBT (2026-06-22)

Use Morningstar to navigate, SEC filings to verify

Apple example · the financials page could not be independently opened in this session

Financials — Income statement → **Annual / TTM** — Pick one basis → **Expand rows** — Read exact labels → **View 10-K** — Trace to filing

Morningstar screen	What you need	Find in the 10-K
Income Statement	Revenue, gross profit, operating income, net income.	Item 8 statements; segment and accounting-policy notes.
Balance Sheet	Cash, investments, current liabilities, current and noncurrent borrowing.	Item 8 balance sheet; debt and securities notes.
Cash Flow	Operating cash flow and the specific cash PP&E line.	Item 8 cash-flow statement, not total investing cash flow.
Debt / maturities	Principal due by year; revolver terms, secured claims and guarantees.	Debt note, liquidity section of Item 7, contractual commitments.
Business / risks	Business model, customers, suppliers, geography and material threats.	Items 1 and 1A; Management's Discussion and Analysis.

Keep units and periods visible

The supplied screenshots state USD millions. 145,097 means US$145.097bn. TTM and annual columns can coincide at a year end; they are not additional periods to add together. "As originally reported" may differ from restated comparisons.

Do not infer the hidden rows

A summary row cannot replace the note. Total investing cash flow includes securities activity and other uses. A debt-schedule total may not be next-year principal. A locked equity star rating is not a bond credit rating.

Access and rights. The Morningstar URL returned an access restriction. The supplied images were used for navigation only; no restricted ratings, licensed analysis, screenshot reproductions or signed S3 access-token links are included.

Sources and dates MORNINGSTAR (User screenshots 2026-11-05) · AAPL (2026-11-03)

Apple’s FY2026 cash generation is large, but definitions still matter

Fiscal year ended 26 September 2026 · Form 10-K filed 3 November 2026

Reported item	USD million	Where / why it matters
Revenue	478,834	Item 8 income statement; denominator for margins.
Gross profit	233,602	Revenue less cost of sales; not cash flow.
Operating income	158,063	Before non-operating items and income tax.
Net income	132,175	Accounting earnings after tax.
Depreciation and amortization	13,423	Cash-flow reconciliation; noncash charge.
Operating cash flow	145,097	Cash generated by operating activities.
Cash PP&E expenditure	9,041	Specific investing line; capex proxy used here.
Cash dividends / buybacks	15,721 / 86,390	Cash distributions, not new borrowing.

Margin calculations

> Gross margin = 233,602 / 478,834 = 48.79%
> Operating margin = 158,063 / 478,834 = 33.01%

Margins measure earnings per dollar of sales. They do not show when tax, suppliers, investment or debt principal must be paid.

EBIT is not EBITDA

> Operating EBITDA proxy
> = 158,063 + 13,423 = US$171,486m

This is an explicitly defined proxy. It is not agency-adjusted or covenant-defined EBITDA. The amount is not free cash available for bondholders.

In everyday terms

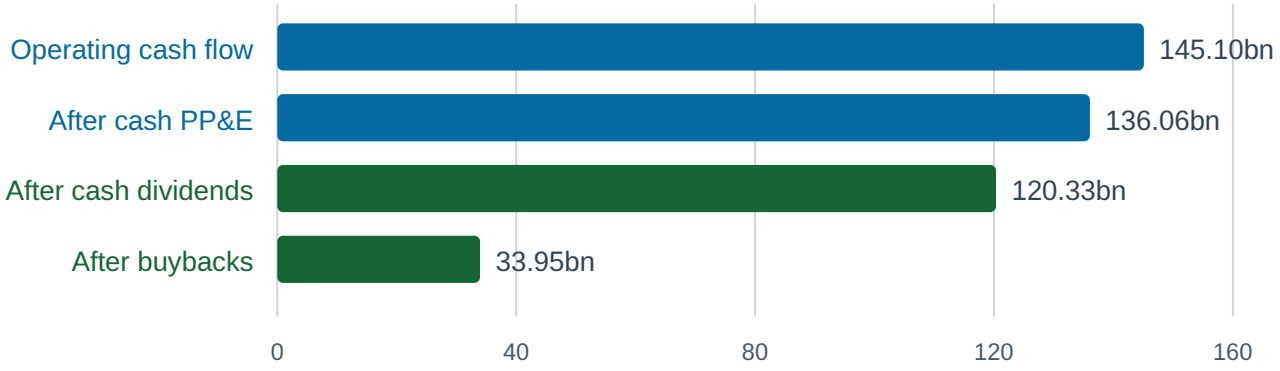
Apple’s reported profits and cash generation provide evidence of capacity. The credit question is how much of that cash is durable and how much is already committed.

Sources and dates AAPL (2026-11-03)

REPORTED INPUTS + CALCULATED BRIDGE

Apple retains US$33.945bn after the selected cash uses

The bridge is a calculation, not a claim that this equals ending cash



How to read. Each bar is the remaining annual cash flow after the named cumulative uses. The final bar excludes principal repayment and other cash uses.
Period / units. Apple FY2026; US$bn. Exact source values are retained in the workbook.

Sources and dates AAPL (2026-11-03)

145,097 − 9,041 = 136,056
136,056 − 15,721 − 86,390 = 33,945 (USD million)

Cash distributions are a management choice, but still use cash. Reducing buybacks can preserve liquidity; it is not proof management will do so. A debt investor should assess financial policy alongside business strength.

The bridge does not subtract interest a second time. Reported operating cash flow already reflects cash-flow accounting for operating items. Do not remove an interest amount again without explicitly rebuilding the cash-flow definition.

In everyday terms
The till generated cash. Buying equipment and returning money to shareholders used some of it. The remaining subtotal still has to face other bills.

Apple can show net debt or net cash, depending on the definition

Report the definition before interpreting the result

Calculation	USD million	Meaning
Commercial paper + current term + noncurrent term	1,997 + 9,906 + 68,208 = 80,111	Book borrowings; not all liabilities.
Book borrowings − cash	80,111 − 36,305 = 43,806	Cash-only net debt.
Less current marketable securities	43,806 − 20,417 = 23,389	Narrower investment-adjusted net debt.
All cash and marketable securities − borrowings	36,305 + 20,417 + 80,213 − 80,111 = 56,824	Broad net cash, not immediate unrestricted cash.
Term principal to carrying amount	78,678 − 277 − 287 = 78,114	Face less discount/costs and hedge adjustment.
Term carrying-value check	9,906 + 68,208 = 78,114	Reconciles to the same amount.

Liquidity has a quality dimension

Securities are not interchangeable with ready cash. Maturity, market value, restrictions, currency and operating location can affect availability. Current ratio = 173,857 / 169,915 = 1.02×, but it does not model cash-flow timing.

Do not manufacture interest coverage

The verified gross-interest input is missing. A sensitivity to a 100bp rate shock measures the change in interest, not total interest expense. Apple's interest-coverage ratio is therefore marked "not verified", rather than inferred.

A maturity schedule must use due principal, not a headline book balance. Note 10 is the starting point. This edition does not claim a complete exact Apple principal ladder from the limited indexed extract.

In everyday terms

A household with savings locked in longer-term investments is not the same as one with the same amount in a current account. Both may be wealthy; only one figure answers "cash today".

Sources and dates AAPL (2026-11-03)

Eight reported annual snapshots cover cash investment; one does not

A consistent first-pass cash screen, not a synchronized ranking of credit quality

Company	Fiscal year end	OCF US$bn	Cash investment US$bn	Surplus US$bn
Apple	2026-09-26	145.097	9.041	**136.056**
Alphabet	2025-12-31	164.713	91.447	**73.266**
Microsoft	2026-06-30	182.935	115.948	**66.987**
Amazon	2025-12-31	139.514	131.819	**7.695**
Meta	2025-12-31	115.800	69.691	**46.109**
NVIDIA	2026-01-25	102.718	6.042	**96.676**
AMD	2025-12-27	6.493	0.974	**5.519**
Micron	2026-09-03	89.675	30.712	**58.963**
Oracle	2026-05-31	31.977	55.663	**-23.686**

Cash-investment surplus = reported OCF − specified cash PP&E / investment line

Oracle's US$23.686bn shortfall is a funding question, not a default conclusion. Cash balances, equity issuance, debt, committed funding, payment timing and future cash generation can bridge a gap. Those sources have different costs and reliability.

Amazon's US$7.695bn gross-cash surplus is not its reported FCF measure. Its company calculation nets US$3.499bn of property proceeds and incentives against cash purchases, producing US$11.194bn. The narrower surplus here retains gross cash investment.

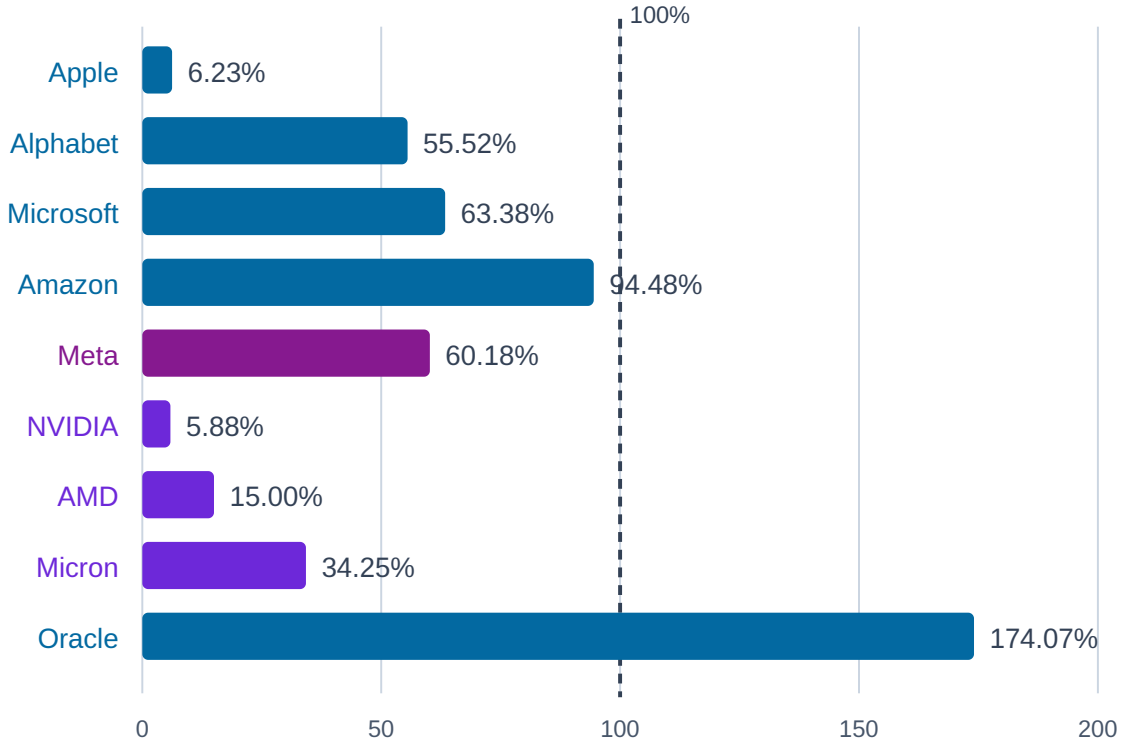
Comparability warning. Fiscal ends run from December 2025 to September 2026. These are annual snapshots, not same-date TTM figures. NVIDIA includes intangibles in its cash investment line; AMD uses continuing operations; finance leases and other cash uses require separate review.

Sources and dates AAPL (2026-11-03) · GOOGL (2026-02-05) · MSFT (2026-07-29) · AMZN (2026-02-06) · META (2026-01-29) · NVDA (2026-02-25) · AMD (2026-02-04) · MU (2026-10-09) · ORCL (2026-06-22)

CALCULATED DIAGNOSTIC

Cash investment absorbs very different shares of operating cash flow

Higher is not automatically worse; the funding plan and returns on investment matter



How to read. A bar above the dashed 100% line means the selected cash investment exceeded OCF in that annual period. Below 100% does not mean every debt, lease, distribution or acquisition is funded.
Period / units. Latest annual periods in the preceding table; %. Company colours follow the approved master.
Limit. Not AI-only capex; reported definitions vary as disclosed.

Sources and dates AAPL (2026-11-03) · GOOGL (2026-02-05) · MSFT (2026-07-29) · AMZN (2026-02-06) · META (2026-01-29) · NVDA (2026-02-25) · AMD (2026-02-04) · MU (2026-10-09) · ORCL (2026-06-22)

Compare capacity with commitments, not with a slogan about AI. Low cash capex can coexist with large purchase commitments or acquisition spending. High cash capex can fund future earnings, but requires credible demand, execution and financing.

In everyday terms

One company can buy equipment from this year’s takings. Another needs savings, lenders or new shareholders. The second is not automatically failing, but its financing plan matters more.

A completed diagnostic scorecard is more useful than a fabricated rating

Reported annual inputs; no arbitrary weights, default probabilities or “safe” labels

Company	Op. margin	Book debt US$bn	Debt less cash US$bn	Surplus / debt	Current ratio	Op. income / interest
Apple	33.01%	80.111	43.806	1.70×	1.02×	Not verified
Alphabet	32.03%	48.543	17.835	1.51×	2.01×	Not verified
Microsoft	46.78%	40.294	19.359	1.66×	1.23×	Not verified
Amazon	11.16%	68.396	-18.414	0.11×	1.05×	Not verified
Meta	41.44%	58.744	22.871	0.78×	2.60×	Not verified
NVIDIA	60.38%	8.468	-2.137	11.42×	3.91×	503.42×
AMD	10.66%	3.222	-2.317	1.71×	2.85×	28.20×
Micron	74.59%	5.179	-33.185	11.39×	3.31×	937.17×
Oracle	30.59%	129.541	98.252	-0.18×	1.12×	4.48×

What the formulas say

> Op. margin = operating income / revenue
> Current ratio = current assets / current liabilities
> Surplus / debt = (OCF − cash investment) / year-end book borrowing
> Coverage = operating income / gross interest expense

What the formulas do not say

Flow-to-stock ratios are not repayment forecasts. A 1.00× surplus/debt ratio does not promise repayment in one year. Existing commitments, business cyclicality, the location of cash and refinancing dates remain material. Negative net debt means cash exceeds this debt definition, not zero risk.

The strongest signals here are cash generation and available balance-sheet liquidity. NVIDIA and AMD show cash exceeding the selected borrowing balance, contrary to the earlier unsupported comparison. Oracle is the clear cash-investment funding-gap case in this annual sample. Neither statement alone ranks bond suitability.

> **Lease and accounting caveat.** Borrowing carrying values are shown as reported. Most debt rows exclude lease liabilities; Micron lease normalization is incomplete. Coverage is shown only where gross interest was verified. Oracle’s 4.48× simple ratio is not its covenant ratio.

Sources and dates AAPL (2026-11-03) · GOOGL (2026-02-05) · MSFT (2026-07-29) · AMZN (2026-02-06) · META (2026-01-29) · NVDA (2026-02-25) · AMD (2026-02-04) · MU (2026-10-09) · ORCL (2026-06-22)

Accounting definitions can change the apparent winner

Reconcile before ranking; do not add R&D, capex and commitments indiscriminately

Issuer / issue	Evidence and treatment	Why the distinction matters
Amazon	Gross cash PP&E 131,819; net cash PP&E 128,320; company FCF 11,194 (USDm).	Property proceeds/incentives change the FCF convention. Gross screen here = 7,695.
Microsoft	Cash PP&E 115,948 (USDm); finance-lease additions and other investing uses are separate.	Cash capex is not total investment, and not necessarily AI-only spending.
NVIDIA	Cash investment 6,042 includes PP&E and intangibles. Other acquisitions/investments excluded.	A fabless model can have large supply commitments despite low cash PP&E.
AMD	OCF 6,493 and cash capex 974 refer to continuing operations (USDm).	Do not add discontinued operations to only one side of the comparison.
Meta	Cash PP&E 69,691 (USDm); finance-lease principal and other property items are separate.	OCF minus PP&E is not automatically Meta's adjusted FCF.
Micron	Annual OCF 89,675; cash PP&E 30,712 before grants (USDm). FY2026 has 53 weeks.	Customer cash deposits of 12,747 are classified in financing, not OCF. Do not "adjust" OCF by subtracting them.
Oracle	FY2026 OCF includes US$4.6bn of customer prepayments with a significant financing component.	OCF quality and timing need review; the disclosure is rounded and not a precise recurring-OCF adjustment here.

R&D spending is not a clean ranking of AI infrastructure investment. Research expense, cash capex, noncash leased assets, acquisitions, power agreements and purchase commitments are different measures. The supplied 20-company R&D ranking is not reproduced because every lead was not verified on one consistent basis.

> **In everyday terms**
> Two people can both say "I spent $10" while one means cash paid and the other means cash plus a credit agreement. A table is only comparable after the definitions are aligned.

Sources and dates AMZN (2026-02-06) · MSFT (2026-07-29) · NVDA (2026-02-25) · AMD (2026-02-04) · META (2026-01-29) · MU (2026-10-09) · ORCL (2026-06-22)

Trace who buys, who builds and who carries the obligations

Core nine issuers have financial inputs; the wider names are a study universe, not a scored ranking

Layer	Names from the supplied study list	Credit questions to test
Cloud, consumer platforms and software	Alphabet, Microsoft, Amazon, Meta, Apple; Oracle, Salesforce, Adobe, IBM, ServiceNow, Palantir, Snowflake, Intuit, Atlassian.	How durable is legacy cash generation? Are AI receipts matching committed construction, equipment and power payments? Apple is primarily consumer devices/services, not interchangeable with cloud infrastructure.
Chip design and architecture	NVIDIA, AMD, Broadcom, Arm, Qualcomm, Marvell, MediaTek; Intel requires separate treatment as an integrated manufacturer/foundry operator.	Customer concentration, technology displacement, export restrictions, supply commitments, inventory and acquisition debt.
Manufacturing and equipment	TSMC; ASML, Applied Materials, Lam Research, KLA, Tokyo Electron. Hon Hai/Foxconn is assembly, not a leading-edge wafer foundry.	Fabrication investment is different from equipment-vendor economics. Test order cycles, geography, customer concentration and cash conversion.
Memory	Micron, Samsung, SK Hynix.	Stress through a memory-price cycle, inventory write-downs and large multiyear fabrication investment.
Servers, networks, power and facilities	Dell, Supermicro, Cisco, Arista, HPE, Lenovo, Vertiv, Equinix.	Do not assume one margin profile: hardware assembly, networking, cooling and data-centre property have different contracts, assets and capital needs.

The study universe is not evidence that every company has the same AI exposure. Outside the nine core financial rows, this edition has not performed a full issuer-by-issuer filing extraction or credit opinion. Company product descriptions and relative claims should be checked before client-specific use.

In everyday terms

The gold rush analogy has limits. The miners, shovel sellers, power suppliers and landlords have different bills, different customers and different assets to recover.

Sources and dates AAPL (2026-11-03) · GOOGL (2026-02-05) · MSFT (2026-07-29) · AMZN (2026-02-06) · META (2026-01-29) · NVDA (2026-02-25) · AMD (2026-02-04) · MU (2026-10-09) · ORCL (2026-06-22)

ANALYSIS + PRIMARY EVIDENCE

AI credit risk follows contracts, not just company logos

BIS Bulletin 140 · 27 October 2026 · official abstract only

Demand
Cloud / model customers

→

Commitments
Capacity, leases, guarantees

→

Vehicles
Asset owners and projects

→

Funding
Bonds, banks, investors

The risk case

A growing share of AI infrastructure is debt-financed. The BIS abstract describes long-term borrowing and off-balance-sheet structures, with commitments from a few large firms supporting vehicle debt and bank exposures. The economic burden can extend beyond the issuer's simple bond balance.

Evidence against a blanket crisis claim

The verified abstract does not establish Treasury funding stress. It reports small and short-lived US Treasury and investment-grade spillovers on AI-financing news days in 2026, while warning that sustained issuance could change the outcome.

Question to ask	Evidence to request
Who must pay if demand disappoints?	Take-or-pay obligations, lease terms, termination rights, guarantees and customer credit.
Is the debt truly non-recourse?	Legal borrower, guarantors, collateral, support agreements and contingent obligations.
Can assets be redeployed?	Alternative tenants, power connection, equipment age, resale cost and technological obsolescence.
Who funds a delay?	Committed liquidity, sponsor support, construction contingency and refinancing assumptions.

This is not a reproduction of the full BIS study. The full PDF and annex could not be retrieved. No event-count, issuance-share, basis-point estimate or chart from inaccessible material is included. The bulletin reflects its authors' analysis, not a guarantee or a policy forecast.

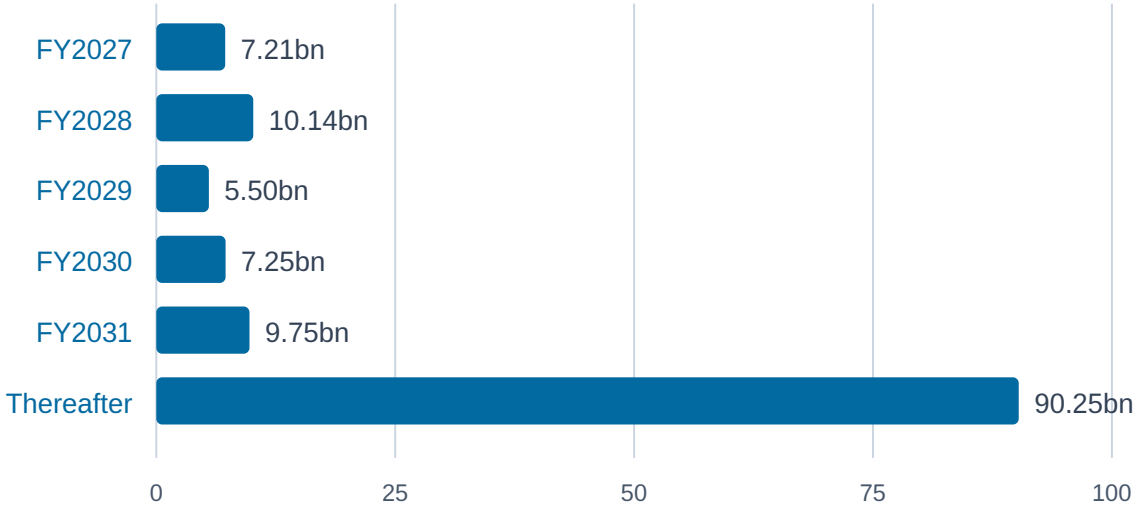
In everyday terms
A lease promise can help someone else borrow. The debt may sit in another company, but the promise can still bring the bill back to the business you are analysing.

Sources and dates BIS (2026-10-27)

ANALYSIS + PRIMARY EVIDENCE

Oracle’s maturity profile is long, but its cash investment still needs funding

Debt-note snapshot at 31 May 2026 · later financing is separate



How to read. Bars show scheduled principal. “Thereafter” aggregates multiple years and is not a single maturity date. Total = US$130.105bn face principal, not US$129.541bn carrying value.
Period / units. Oracle FY2026 debt disclosure, US$bn; filed 22 June 2026.
Limit. Subsequent borrowing, equity proceeds and uses are not rolled into this historical ladder.

Sources and dates ORCL_DEBT (2026-06-22)

Available financing is not free cash

The US$10bn revolving facility was undrawn at year end. Drawing depends on contractual conditions. Its stated covenant uses defined consolidated EBITDA and consolidated net interest expense with a minimum 3.0× ratio; our 4.48× operating-income/gross-interest calculation is not a covenant test.

Funding sources can counter the risk case

A November SEC filing confirms both debt and equity financing. Oracle reported closing US$10bn of notes and US$10bn of common stock on 4 November 2026. Equity issuance and financing access challenge an automatic-default narrative, but do not by themselves establish future investment returns.

In everyday terms

A long mortgage schedule helps, but it does not pay this year’s construction bill. The annual funding gap and the debt calendar are separate questions.

Sources and dates ORCL_CLOSE (2026-11-04)

Original pricing must not masquerade as a current quote

Verified Oracle terms; partial Microsoft extraction; Meta primary terms unresolved

Oracle tranche	Size US$bn	Coupon	Issue price	Issue YTM	UST spread
4 Nov 2029	2.000	7.000%	99.835	7.062%	+240bp
4 Nov 2031	1.750	7.450%	99.942	7.464%	+275bp
2033 year verified	2.250	7.850%	Not retrieved	Not retrieved	Not retrieved
2037 year verified	2.500	8.000%	Not retrieved	Not retrieved	Not retrieved
2046 year verified	1.500	8.650%	Not retrieved	Not retrieved	Not retrieved

Oracle's trade date is 2 November; the closing is confirmed on 4 November. The 2029 note uses a 4.875% Treasury due 15 October 2029 with a 4.662% benchmark yield. The 2031 note uses a 4.750% Treasury due 31 October 2031 with a 4.714% yield. Prices are percentages of principal. These are original terms, not today's yields.

> 2029: 7.062% − 4.662% = 2.400% = 240bp
> 2031: 7.464% − 4.714% = 2.750% = 275bp

Microsoft: incomplete fields remain incomplete

The primary header identifies ten series, including two floating-rate notes. The 29 October term sheet states 4 November settlement. Eight fixed coupons span 4.850% to 6.350%; full sizes, yields and spreads were not retrieved here. This is not a complete verified US$23bn comparison.

Meta: do not promote a lead into proof

The supplied USD terms were not verified directly from the primary filing. They are withheld from the verified numerical table. A later EUR offering is a separate instrument set. Neither original USD yields nor any current secondary quote is inferred.

In everyday terms

A bond's launch menu is useful evidence, but it is not a live price board. Empty verified fields are preferable to confident-looking guesses.

Sources and dates ORCL_FWP (2026-11-02) · ORCL_CLOSE (2026-11-04) · MSFT_FWP (2026-10-29)

ANALYSIS + PRIMARY EVIDENCE

A wider spread is meaningful only after the comparison is controlled

A bond's yield, its credit spread and its default probability are different quantities

Benchmark yield
Base curve for the currency

→

+ Spread
Credit, liquidity and risk pricing

→

Bond yield
With options and convention caveats

Match or explain	Why it changes the comparison
Currency and benchmark	USD Treasury, swap and other currency curves are not interchangeable. A Treasury par yield is not every individual benchmark bond.
Date, timestamp and quote side	Different sessions, bid/ask prices, accrued interest or settlement assumptions create artificial differences.
Tenor and rate / spread duration	A long bond typically carries different sensitivity and future uncertainty from a short bond.
Issuer entity, rank and collateral	Parent, subsidiary, guaranteed, secured, unsecured and subordinated claims can have different expected losses.
Embedded options and covenants	Callable, puttable, perpetual and reset securities require the relevant valuation convention, often model-dependent OAS.
Liquidity, issue size and technical supply	New-issue concessions, trading activity and investor demand affect spreads without an equivalent change in default risk.

Simple illustration, not an issuer estimate

Spread = 200bp = 0.02
Assumed recovery = 40%; LGD = 60%
Hazard ≈ 0.02 / 0.60 = 3.33% per year
5-year model PD = 1 − exp(−0.033333 × 5)
= 15.35%

Why that is not a default forecast

This is a simplified risk-neutral model calculation. It assumes constant hazard and recovery conventions and ignores liquidity and other premia. A physical probability of default requires further modelling and calibration. The same observed spread can support different interpretations.

In everyday terms
Comparing bond spreads without matching the instruments is like comparing insurance premiums without checking what is insured or the deductible.

Sources and dates FINRA_SPREAD (Page checked 2026-11-06) · NYFED (2026-02-24)

A solvent issuer can still produce a negative bond return

Hypothetical examples separate cash repayment, market price and home-currency returns

Rates: price falls without a default

A five-year 5% annual-coupon bond priced at a 5% yield is worth 100. At an instantaneous 6% yield, the same contractual cash flows are worth 95.7876, a 4.21% price loss. No time passage, coupon receipt or default is assumed.

```
P(y) = Σ[t=1…5] 5 / (1+y)^t
+ 100 / (1+y)^5
```

FX: a USD gain can shrink in SGD

A 5% USD return is only 1.11% in SGD in this example. Start FX = 1.3500 SGD per USD; end FX = 1.3000 SGD per USD. These are invented teaching inputs, not current market quotations.

```
SGD return = (1 + 5%) × 1.3000 / 1.3500 − 1
= 1.1111%
```

Duration is a sensitivity, not a credit rating

```
Separate example: modified duration = 7 years
Yield rises 100bp: first-order ΔP/P ≈ −7 × 0.01 = −7%
```

The approximation ignores convexity and assumes a small parallel yield move. The seven-year duration is a separate input, not the measured duration of the five-year bond above. For a corporate bond, benchmark yields and spreads can move in different directions.

Component	What to keep separate
Coupon	Contractual cash paid on principal.
Current yield	Annual coupon divided by current price; ignores redemption gain/loss.
YTM / yield-to-worst	Valuation measures subject to the relevant cash-flow and option assumptions.
Total return	Coupons + price change + reinvestment effects, after fees and FX as applicable.

In everyday terms

Being paid as promised does not protect the price you receive when selling early, or the amount you receive after changing currencies.

Sources and dates FINRA_YIELD (Page checked 2026-11-06) · SEC_BOND (Undated; checked 2026-11-06)

HYPOTHETICAL + FORMULA AUDIT

The corrected waterfall pays both claims in full at US$100

Simple single-pool illustration; real legal structures and recovery timing can change outcomes

Scenario	Gross assets	Costs	Distributable	Senior paid / 80	Junior paid / 20	Equity
Full payment	$100	$0	$100	$80 / 100%	$20 / 100%	$0
Costs first	$100	$10	$90	$80 / 100%	$10 / 50%	$0
Asset loss + costs	$70	$10	$60	$60 / 75%	$0 / 0%	$0
Value above debt	$120	$0	$120	$80 / 100%	$20 / 100%	$20

A = max(gross realized assets − paid costs, 0)
Senior paid = min(senior claim, A)
Junior paid = min(junior claim, max(A − senior paid, 0))
Equity residual = max(A − senior paid − junior paid, 0)
Recovery rate = amount paid / claim; LGD = 1 − recovery rate

If US$100 is already distributable, costs have already been dealt with. Do not subtract them again. In the first row, 100 = 80 + 20 and both creditors recover their entire claims. In the second, 100 = 10 costs + 80 senior + 10 junior. All rows conserve value and cap payments at the claim.

How analysts estimate the pool

Book assets are not liquidation proceeds. A going-concern valuation may use stressed cash generation and a valuation multiple, or an asset sale may use realizable cash values. Deduct costs and claims at the correct legal entity, then allocate by enforceable priorities.

Why actual recovery varies

Collateral and seniority are not universal shortcuts. Jurisdiction, administration costs, liens, guarantees, subsidiary claims, accrued interest and timing matter. Senior unsecured bonds can still lose money. Equity ranks last in this simple model.

In everyday terms

There is no junior loss when the available pot exactly covers both debts. The loss begins only when costs or a smaller pot leave less than the total claims.

Sources and dates SEC_PRIORITY (Undated; checked 2026-11-06) · SEC_BOND (Undated; checked 2026-11-06)

Interest coverage can remain positive while liquidity fails

A hypothetical issuer, not company guidance or a forecast

Annual USDm unless ratio	Base	Sales −10%	Margin −4pp	Capex +50%	Rate +4pp	Combined
Operating income	20.00	18.00	16.00	20.00	20.00	14.40
Cash interest	2.00	2.00	2.00	2.00	2.80	2.80
OCF proxy	19.40	17.80	16.20	19.40	18.76	14.28
FCF proxy	9.40	7.80	6.20	4.40	8.76	-0.72
Closing cash before new funding	7.40	5.80	4.20	2.40	6.76	-2.72
Funding to restore $5 minimum	0.00	0.00	0.80	2.60	0.00	7.72
Operating income / interest	10.00×	9.00×	8.00×	10.00×	7.14×	5.14×

Operating income = revenue × operating margin
Interest = repriced debt × new rate + other debt × its rate
Cash tax = max(operating income − interest, 0) × tax rate
OCF proxy = operating income + D&A − interest − tax − working-capital use
FCF = OCF proxy − capex
Closing cash = opening cash + FCF − unrolled principal − dividends − buybacks

The base case starts with US$100m revenue, 20% operating margin and US$10m cash. D&A is 5, tax is 20%, capex is 10, dividends are 2 and total debt is 50. Of that debt, 20 rolls at the start of the year; 30 pays 4% and 10 is repaid at year end. No working-capital use, buybacks or new rescue funding is assumed.

The combined case still covers annual interest 5.14× but needs US$7.72m of funding. Its revenue is 90, margin 16%, capex 15 and refinancing rate 8%. Closing cash is negative 2.72; restoring the minimum cash buffer of 5 requires 7.72. Stopping the dividend helps by 2, but does not eliminate the gap.

In everyday terms
Earning enough to pay interest is not the same as having enough cash to repay principal, build facilities and keep the business operating.

Use three separate decisions before buying a bond

Issuer resilience → bond protection → price and portfolio suitability

1 · Can the issuer pay?

Test sustainable cash generation under stress. Read business risks, cash conversion, leases, commitments, debt maturities and committed liquidity.

2 · What does this bond own?

Identify the legal claim. Issuer entity, guarantees, collateral, ranking, covenants, call terms, currency and settlement all matter.

3 · Is the price worthwhile?

Compare matched risks and your holding needs. Check executable price, YTM/YTW, duration, OAS where appropriate, fees, FX and concentration.

Working conclusion from the verified sample	What could change it
Apple shows large annual cash-generation capacity and broad securities resources.	Weaker operating performance, less accessible liquidity, large commitments, aggressive distributions or changed debt terms.
Microsoft and Alphabet cover the selected annual cash investment from OCF.	Faster cash and lease investment, weaker monetisation, working-capital changes or later balance-sheet funding.
NVIDIA and AMD hold cash above the selected borrowing balance.	A downturn in demand, customer concentration, supply commitments, acquisition spending or changing cash quality.
Oracle requires greater attention to financing and investment timing in this snapshot.	Equity proceeds, committed financing and durable customer cash flows can offset risk; project delays, spending overruns or financing constraints can intensify it.
No bond is declared cheap or suitable from these annual accounts alone.	A dated executable quote and matched instrument comparison are required for a relative-value view.

Monitor the evidence that can break the thesis. The practical review list is quarterly cash flow and capex, unrestricted liquidity, debt due within 12–24 months, lease and purchase commitments, covenant headroom using the actual definitions, and distributions to shareholders. A watchlist is not an automated task created by this report.

> **In everyday terms**
> A strong company is the starting point. A good bond also needs a sound contract, a sensible price and a place in the investor's portfolio.

Sources and dates AAPL (2026-11-03) · MSFT (2026-07-29) · GOOGL (2026-02-05) · NVDA (2026-02-25) · AMD (2026-02-04) · ORCL (2026-06-22) · SEC_BOND (Undated; checked 2026-11-06)

What was checked, what remains open

Methodology, access boundaries and publication safeguards

Area	Treatment
Data cutoff	Research checked on 6 Nov 2026, Singapore time. Financial observations retain their actual fiscal periods; no claim of a synchronized market close.
Primary evidence	Nine SEC annual filings and original disclosure extracts; exact filing/access notes in the source ledger. Some full FY2026 bodies were unavailable, so indexed SEC extracts support the cited numbers.
Calculations	Source precision retained in Excel; display rounding only. All financial diagnostics, waterfalls, scenarios and valuation examples use transparent formulas.
Missing information	Missing ratings, gross interest inputs and issue-pricing fields remain unavailable. No full current peer ranking or complete latest issuance inventory is claimed.
Visual reference	The DBS/STI page could not be loaded and its repository path returned 404. Its exact layout was not inspected. The attached V1.2 Excel and PDF, plus the described card approach, govern this edition.
BIS and Morningstar	BIS official abstract only; no full-PDF analysis claimed. Morningstar access restricted; user screenshots guide navigation only.
Market charts	No synchronized executable bond, FX or SPX dataset was verified. Only relevant annual financial charts and explicitly hypothetical price/FX examples are included.
Publication and privacy	Local original text, derived visuals and linked sources. No copied proprietary ratings, screenshots, client data, signed token URLs, tracking or external assets. No website, original attachment or scheduled task was changed.

This is educational research, not an assurance of repayment or a personal recommendation. The stated methods do not replace legal review of an instrument, a current market quote, suitability assessment or refreshed issuer analysis. Public disclosures may be corrected or superseded; source links are retained so readers can check.

Reusing the local article

The HTML is self-contained and not published. It contains no external fonts, image downloads or analytics. Before public reuse, confirm any required employer/compliance approval and licensing for any later third-party additions. Links to evidence do not transfer rights to copy restricted material.

Sources and dates STYLE (2026-11-05) · REF_LAYOUT (2026-11-06 URL) · MORNINGSTAR (User screenshots 2026-11-05) · BIS (2026-10-27)

Sources and evidence trail 1

Publication date is not the same as the period measured

[AAPL] Apple FY2026 Form 10-K

Filed / published: 2026-11-03 · **Observation:** 2026-09-26
Locator: Item 7 liquidity; Item 8 statements; Note 10 Debt
Access / limits: Indexed SEC primary extracts; direct full HTML retrieval unavailable
Open primary source / reference

[GOOGL] Alphabet FY2025 Form 10-K

Filed / published: 2026-02-05 · **Observation:** 2025-12-31
Locator: Item 8 consolidated statements; debt and securities disclosures
Access / limits: Full SEC HTML
Open primary source / reference

[MSFT] Microsoft FY2026 Form 10-K / official annual report

Filed / published: 2026-07-29 · **Observation:** 2026-06-30
Locator: Item 8 statements and debt note; official annual report statements
Access / limits: SEC/issuer indexed primary extracts; full SEC HTML exceeded retrieval limit
Open primary source / reference

[AMZN] Amazon FY2025 Form 10-K

Filed / published: 2026-02-06 · **Observation:** 2025-12-31
Locator: Item 8 statements; Note 4 Leases; Note 6 Debt; Item 7 FCF reconciliation
Access / limits: Full SEC HTML
Open primary source / reference

[META] Meta FY2025 Form 10-K

Filed / published: 2026-01-29 · **Observation:** 2025-12-31
Locator: Item 8 consolidated financial statements; debt disclosure
Access / limits: Full SEC HTML
Open primary source / reference

[NVDA] NVIDIA FY2026 Form 10-K

Filed / published: 2026-02-25 · **Observation:** 2026-01-25
Locator: Item 8: p.51 income, p.53 balance, p.55 cash flow; commitments
Access / limits: Full SEC HTML
Open primary source / reference

[AMD] AMD FY2025 Form 10-K

Filed / published: 2026-02-04 · **Observation:** 2025-12-27
Locator: Item 8: p.58 income, p.60 balance, p.62 cash flows
Access / limits: Full SEC HTML; continuing operations cash-flow basis
Open primary source / reference

[MU] Micron FY2026 Form 10-K

Filed / published: 2026-10-09 · **Observation:** 2026-09-03
Locator: Items 7–8; Note 4 Cash and Investments; Note 9 Debt; Note 16 customer deposits
Access / limits: Indexed SEC primary extracts; fiscal year has 53 weeks
Open primary source / reference

Sources and evidence trail 2

Publication date is not the same as the period measured

[ORCL] Oracle FY2026 Form 10-K

Filed / published: 2026-06-22 · **Observation:** 2026-05-31
Locator: Item 8 statements; Note 6 Notes Payable and Other Borrowings
Access / limits: Full filing oversized; original SEC XBRL statement/Note 6 pages retrieved
Open primary source / reference

[MSFT_IR] Microsoft official FY2026 annual report

Filed / published: 2026 annual report · **Observation:** 2026-06-30
Locator: Income Statements, Balance Sheets, Cash Flows, debt note
Access / limits: Indexed official issuer extracts; direct page returned 404
Open primary source / reference

[ORCL_BS] Oracle SEC XBRL balance sheet

Filed / published: 2026-06-22 · **Observation:** 2026-05-31
Locator: Consolidated Balance Sheets
Access / limits: Full SEC page
Open primary source / reference

[ORCL_IS] Oracle SEC XBRL income statement

Filed / published: 2026-06-22 · **Observation:** 2026-05-31
Locator: Consolidated Statements of Operations
Access / limits: Full SEC page
Open primary source / reference

[ORCL_DEBT] Oracle SEC XBRL debt note

Filed / published: 2026-06-22 · **Observation:** 2026-05-31
Locator: Note 6; debt maturity table; revolving facility covenant
Access / limits: Full SEC page
Open primary source / reference

[BIS] BIS Bulletin 140: Building AI on credit: structures and spillovers

Filed / published: 2026-10-27 · **Observation:** 2026 study
Locator: Official abstract: three key takeaways
Access / limits: Abstract only. Full eight-page PDF and online annex not retrieved
Open primary source / reference

[SEC_BOND] SEC Investor.gov: Corporate Bonds

Filed / published: Undated; checked 2026-11-06 · **Observation:** General educational guidance
Locator: Corporate bonds; bankruptcy; basic risks
Access / limits: Primary regulator guidance
Open primary source / reference

[SEC_PRIORITY] SEC Investor.gov: Bankruptcy for a Public Company

Filed / published: Undated; checked 2026-11-06 · **Observation:** General educational guidance
Locator: Priority of creditors over equity
Access / limits: Primary regulator guidance; not case-specific legal advice
Open primary source / reference

SOURCE LEDGER

Sources and evidence trail 3

Publication date is not the same as the period measured

[SEC_RATING] SEC: The ABCs of Credit Ratings

Filed / published: 2017-10-12 · **Observation:** General educational guidance
Locator: What a credit rating is and is not
Access / limits: Primary regulator guidance; no issuer ratings assigned in report
Open primary source / reference

[FINRA_SPREAD] FINRA: What You Need to Know About Bond Spreads

Filed / published: Page checked 2026-11-06 · **Observation:** General educational guidance
Locator: Benchmark and spread interpretation
Access / limits: Primary regulator educational page
Open primary source / reference

[FINRA_YIELD] FINRA: Bond Yield and Return

Filed / published: Page checked 2026-11-06 · **Observation:** General educational guidance
Locator: Yield versus realized return
Access / limits: Primary regulator educational page
Open primary source / reference

[NYFED] NY Fed: Estimating the Term Structure of Corporate Bond Risk Premia

Filed / published: 2026-02-24 · **Observation:** Study sample 2002–2020
Locator: Extracting Risk Premia from Yields; model limitations
Access / limits: Primary research commentary; author views; not current spread observations
Open primary source / reference

[MSFT_FWP] Microsoft pricing term sheet

Filed / published: 2026-10-29 · **Observation:** Trade 2026-10-29; stated settlement 2026-11-04
Locator: Cover; securities; trade/settlement dates; floating principal amounts
Access / limits: Partial indexed SEC extracts; fixed tranche sizes/prices/yields not independently verified
Open primary source / reference

[ORCL_FWP] Oracle final pricing term sheet

Filed / published: 2026-11-02 · **Observation:** Trade 2026-11-02; settlement 2026-11-04
Locator: Five fixed tranches; full pricing fields retrieved for 2029 and 2031
Access / limits: Indexed SEC primary extracts; remaining price/yield fields unavailable
Open primary source / reference

[ORCL_CLOSE] Oracle Form 8-K

Filed / published: 2026-11-04 · **Observation:** Closing 2026-11-04
Locator: Item 8.01: completed $10bn notes and $10bn common stock offerings
Access / limits: Indexed SEC primary extract; closing confirmed for Oracle
Open primary source / reference

[MORNINGSTAR] Apple Morningstar financials

Filed / published: User screenshots 2026-11-05 · **Observation:** Screenshot rating timestamp 2026-11-04
Locator: Income Statement / Balance Sheet / Cash Flow navigation
Access / limits: Direct access 403. No paywalled data or ratings reproduced
Open primary source / reference

SOURCE LEDGER

Sources and evidence trail 4

Publication date is not the same as the period measured

[STYLE] Colour & Tagging Standard V1.2 Merged

Filed / published: 2026-11-05 · **Observation:** Approved supplied standard
Locator: Excel canonical mappings; PDF chart/layout/accessibility/navigation rules
Access / limits: Both supplied files inspected; sources unchanged
Local user attachments: Market_Colour_Tag_Master_v1_2_Merged(2).xlsx; Colour_Tag_Visual_Design_Guide_v1_2_Merged(2).pdf

[REF_LAYOUT] DBS/STI visual reference

Filed / published: 2026-11-06 URL · **Observation:** Design reference only
Locator: Requested reference page
Access / limits: Could not load; repository file lookup also returned 404. Exact layout not inspected
Open primary source / reference